```
clear;
clc;
close all;

Fs = 8000;
fc = 1500;
wc = fc/(Fs/2);
N = 50;

% Window Functions
w_rect = rectwin(N+1);
w_hamm = hamming(N+1);
w_hann = hann(N+1);
w_black = blackman(N+1);

% FIR Filter Design
b_rect = fir1(N,wc,w_rect);
b_hamm = fir1(N,wc,w_hamm);
b_hann = fir1(N,wc,w_hann);
b_black = fir1(N,wc,w_black);

% Frequency Response
[H_rect,f] = freqz(b_rect,1,1024,Fs);
[H_hamm,~] = freqz(b_hamm,1,1024,Fs);
[H_hann,~] = freqz(b_hann,1,1024,Fs);
[H_black,~] = freqz(b_black,1,1024,Fs);

% Magnitude in dB
M_rect = 20*log10(max(abs(H_rect),1e-10));
M_hamm = 20*log10(max(abs(H_hamm),1e-10));
M_hann = 20*log10(max(abs(H_hann),1e-10));
M_black = 20*log10(max(abs(H_black),1e-10));

% Display Numerical Outputs
fprintf('=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===\n\n');
```

```
=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===
```

```
fprintf('Filter Order (N): %d\n',N);
```

```
Filter Order (N): 50
```

```
fprintf('Cutoff Frequency: %.2f Hz\n\n',fc);
```

```
Cutoff Frequency: 1500.00 Hz
```

```
fprintf('--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_rect(1:5));
```

```
   -0.0118         0    0.0128    0.0103   -0.0058
```

```
fprintf('--- HAMMING WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- HAMMING WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_hamm(1:5));
```

```
   -0.0009         0    0.0012    0.0011   -0.0008
```

```
fprintf('--- HANNING WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- HANNING WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_hann(1:5));
```

```
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588
```

```
fprintf('--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---\n');
```

```
--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---
```

```
disp(b_black(1:5));
```

```
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```

```
% Plot Magnitude Responses
figure('Name','FIR Windowing Techniques Comparison');

subplot(2,1,1);
plot(f,M_rect,'k','LineWidth',1);
hold on;
plot(f,M_hamm,'b','LineWidth',1.5);
plot(f,M_hann,'g','LineWidth',1.5);
plot(f,M_black,'r','LineWidth',1.5);

grid on;
legend('Rectangular','Hamming','Hanning','Blackman');
title('FIR Low-Pass Filter Magnitude Response (N = 50)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-100 10]);

% Phase Response
subplot(2,1,2);
```

```
plot(f,unwrap(angle(H_hamm))*180/pi,'b','LineWidth',1.5);
grid on;
title('Phase Response (Hamming Window)');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

